

Fractals and their function and clause

Mandelbrot

$$I(n+1, c) = I(n, c)^2 + c$$
$$I(0, c) = c$$

The series should stay bounded.

Mandelbrot Cubed

$$I(n+1, c) = I(n, c)^3 + c$$
$$I(0, c) = c$$

The series should stay bounded

Julia

$$I(n+1, c) = \cos(I(n, c))$$
$$I(0, c) = c$$

The series should stay bounded $\Leftrightarrow$ the series should approach the Dottie number

Spirals

$$I(n+1, c) = \ln(1 + z^2)$$
$$I(0, c) = c$$

The should converge to a number with an absolute value smaller or equal to one.

Crab

$$I(n+1, c) = z^{1-z}$$
$$I(0, c) = c$$

The should converge to a number with an absolute value smaller or equal to one.

Singularity

$$I(n+1, c) = e^{c^2+z^2}$$
$$I(0, c) = c$$

The should converge to a number with an absolute value smaller or equal to one.

Cubed Singularity

$$I(n+1, c) = e^{c^3+z^2}$$
$$I(0, c) = c$$

The should converge to a number with an absolute value smaller or equal to one.